NAME : ANJALI GUPTA	ROLL NO: S010
DATE OF ASSIGNMENT:	DATE OF SUBMISSION:

PLANT NURSING MANAGEMENT SYSTEM

CODE:

```
plants = ["Rose", "Tulsi", "Money Plant", "Aloe Vera", "Jasmine"]
prices = [150, 80, 200, 120, 100]

# Linked List
class Node:
    def __init__(self, name, price):
        self.name = name
        self.price = price
        self.next = None

head = None

for i in range(len(plants)):
    new = Node(plants[i], prices[i])
    new.next = head
    head = new
```

```
# Queue
queue = []

while True:
    print("\n===== PLANT NURSERY MANAGEMENT =====")
    print("1. Display All Plants")
    print("2. Sort Plants by Price")
    print("3. Search Plant")
    print("4. Add Customer to Queue")
    print("5. Serve Customer")
    print("6. Exit")

    choice = int(input("Enter your choice: "))

    # Display All
    if choice == 1:
        print("\n--- All Plants ---")
        temp = head

        while temp:
            print(temp.name, "₹", temp.price)
            temp = temp.next

        print("\n--- Customer Queue ---")
        if queue:
            print(queue)
        else:
            print("No customers in queue.")
```

```python
    # Bubble Sort
    elif choice == 2:
        for i in range(len(prices)):
            for j in range(len(prices) - i - 1):
                if prices[j] > prices[j + 1]:
                    prices[j], prices[j + 1] = prices[j + 1], prices[
                    plants[j], plants[j + 1] = plants[j + 1], plants[

        print("\n--- Plants Sorted by Price ---")
        for i in range(len(plants)):
            print(plants[i], "₹", prices[i])

    # Searching
    elif choice == 3:
        search = input("Enter plant name: ")

        if search in plants:
            index = plants.index(search)
            print("Plant Found! Price = ₹", prices[index])
        else:
            print("Plant Not Found")

    # Queue - Add Customer
    elif choice == 4:
        name = input("Enter customer name: ")
        queue.append(name)
        print(name, "added to queue.")

    # Queue - Serve Customer
    elif choice == 5:
        if queue:
            print("Serving:", queue.pop(0))
        else:
            print("Queue is empty.")
```

```python
    # Exit
    elif choice == 6:
        print("Thank you!")
        break

    else:
        print("Invalid choice!")
```

OUTPUT:

```
===== PLANT NURSERY MANAGEMENT =====
1. Display All Plants
2. Sort Plants by Price
3. Search Plant
4. Add Customer to Queue
5. Serve Customer
6. Exit
Enter your choice: 1

--- All Plants ---
Jasmine ₹ 100
Aloe Vera ₹ 120
Money Plant ₹ 200
Tulsi ₹ 80
Rose ₹ 150

--- Customer Queue ---
No customers in queue.

===== PLANT NURSERY MANAGEMENT =====
1. Display All Plants
2. Sort Plants by Price
3. Search Plant
4. Add Customer to Queue
5. Serve Customer
6. Exit
Enter your choice: 2
```

```
--- Plants Sorted by Price ---
Tulsi ₹ 80
Jasmine ₹ 100
Aloe Vera ₹ 120
Rose ₹ 150
Money Plant ₹ 200

===== PLANT NURSERY MANAGEMENT =====
1. Display All Plants
2. Sort Plants by Price
3. Search Plant
4. Add Customer to Queue
5. Serve Customer
6. Exit
Enter your choice: 3
Enter plant name: mogra
Plant Not Found

===== PLANT NURSERY MANAGEMENT =====
1. Display All Plants
2. Sort Plants by Price
3. Search Plant
4. Add Customer to Queue
5. Serve Customer
6. Exit
Enter your choice: 4
Enter customer name: xyz
xyz added to queue.
```

```
===== PLANT NURSERY MANAGEMENT =====
1. Display All Plants
2. Sort Plants by Price
3. Search Plant
4. Add Customer to Queue
5. Serve Customer
6. Exit
Enter your choice: 4
Enter customer name: xyz
xyz added to queue.

===== PLANT NURSERY MANAGEMENT =====
1. Display All Plants
2. Sort Plants by Price
3. Search Plant
4. Add Customer to Queue
5. Serve Customer
6. Exit
Enter your choice: 5
Serving: xyz

===== PLANT NURSERY MANAGEMENT =====
1. Display All Plants
2. Sort Plants by Price
3. Search Plant
4. Add Customer to Queue
5. Serve Customer
6. Exit
Enter your choice: 6
Thank you!
```